



SONA COLLEGE OF TECHNOLOGY

AUTONOMOUS

**Department of Electronics and Communication
Engineering**

U23EC301 SIGNALS AND SYSTEMS LAB MANUAL

SONA COLLEGE OF TECHNOLOGY

Junction Main Road, Sona Nagar,
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Name :

Branch :

Semester :

Register No. :

Ceritifed that this is the Bona-fide Record of work done by the above student in during the year

Lab in-Charge

Head of the Department

Submitted for the University Practical Examination held on

Internal Examiner

External Examiner

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Syllabus

List of Experiments

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- a. To study various MATLAB commands for creating two- and three-dimensional plots.
- b. Write a MATLAB program to plot the following continuous time and discrete time signals.
 - i. Step Function
 - ii. Impulse Function
 - iii. Exponential Function
 - iv. Ramp Function
 - v. Sine Function

2 Time and Amplitude transformations

Write a MATLAB program to perform amplitude-scaling, timescale and time shifting on a given signal.

3. Convolution of given signals

Write a MATLAB program to obtain linear convolution of the given sequences.

4. Impulse response and Step response of a given system

- a. Write a MATLAB program to find the impulse response and step response of a system from its difference equation.
- b. Compute and plot the response of a given system to a given input.

5. Fourier Series and Gibbs Phenomenon

- a. To calculate Fourier series coefficients associated with Square Wave.
- b. To Sum the first 10 terms and plot the Fourier series as a function of time.
- c. To Sum the first 50 terms and plot the Fourier series as a function of time

6. Calculating transforms using MATLAB

Calculate and plot Fourier transform of a given signal.

7. Calculating transforms using MATLAB

Calculate and plot Z-transform of a given signal.

8. Pole-zero diagram and ROC

Write a MATLAB program to find pole-zero diagram from the given system function.

9. Frequency response of a system

Write a MATLAB program to plot magnitude and phase response of a given system.

10. Checking linearity/non-linearity of a system using SIMULINK

- a. Build a system that amplifies a sine wave by a factor of two.
- b. Test the linearity of this system using SIMULINK.

LAB EXPERIMENT 1

AIM: -

1. To plot the following continuous time and discrete time signals.
 - i. Step Function
 - ii. Impulse Function
 - iii. Exponential Function
 - iv. Ramp Function
 - v. Sine Function
2. To study various MATLAB commands for creating two- and three-dimensional plots.

SOFTWARE USED: - MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and run the program
- For the output see command window\ Figure window

PROGRAM:

CONTINUOUS TIME SIGNALS

```
t=(-2:0.01:10);  
impulse = t==0;  
plot(t, impulse)  
xlabel('Time')  
ylabel('Amplitude')  
title('Impulse Function')  
  
unitstep = t>=0;  
plot(t, unitstep)  
xlabel('Time')  
ylabel('Amplitude')  
title('Unit Step Function')
```

```

t=(0:0.01:10);
step = t>=0;
Exponential=exp(t).*step;
plot(t, Exponential)
xlabel('Time')
ylabel('Amplitude')
title('Growing Exponential Function')

Exponential=exp(-t).*step;
plot(t, Exponential)
xlabel('Time')
ylabel('Amplitude')
title('Decaying Exponential Function')

ramp = t.*step;
plot(t,ramp)
xlabel('Time')
ylabel('Amplitude')
title('Ramp Function')

x=(0:pi/10:2*pi);
y=sin(x)';
plot(x,y)
title('Sine Signal')
xlabel('Time')
ylabel('Amplitude')

```

DISCRETE TIME SIGNALS

```

t=(-2:1:10);
impulse = t==0;
stem(t, impulse)
xlabel('Time')
ylabel('Amplitude')
title('Impulse Function')

unitstep = t>=0;
stem(t, unitstep)

```



```

xlabel('Time')
ylabel('Amplitude')
title('Unit Step Function')

t=(0:1:10);
step = t>=0;
Exponential=exp(t).*step;
stem(t, Exponential)
xlabel('Time')
ylabel('Amplitude')
title('Growing Exponential Function')

Exponential=exp(-t).*step;
stem(t, Exponential)
xlabel('Time')
ylabel('Amplitude')
title('Decaying Exponential Function')

ramp = t.*step;
stem(t,ramp)
xlabel('Time')
ylabel('Amplitude')
title('Ramp Function')

x=(0:pi/10:2*pi);
y=sin(x)';
stem(x,y)
title('Sine Signal')
xlabel('Time')
ylabel('Amplitude')

```

2D Plot

```

clc
clear all
x=(0:pi/10:2*pi);
y=sin(x)';
plot(x,y)

```

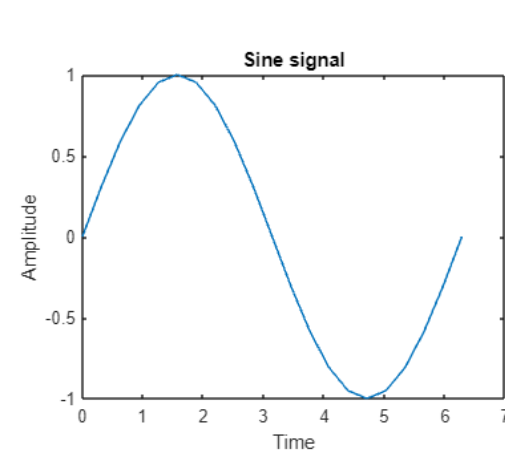
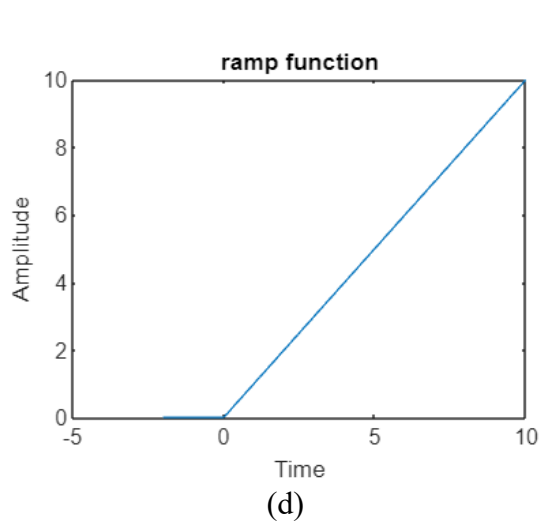
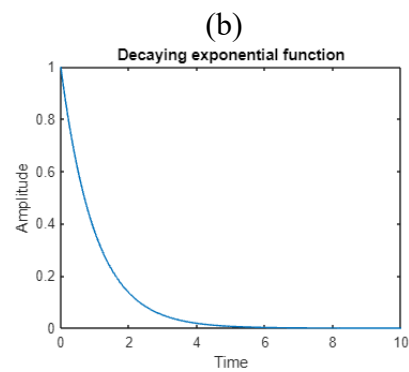
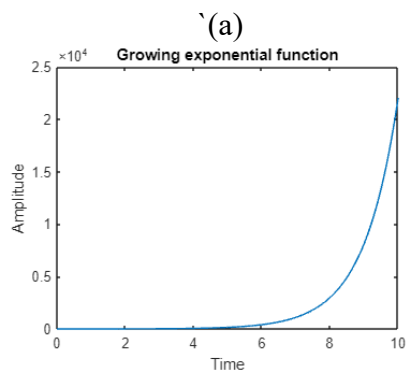
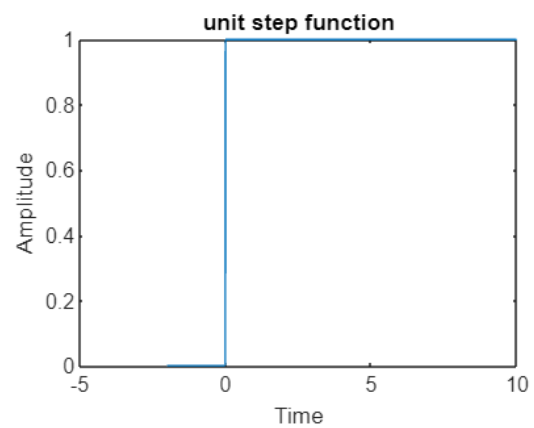
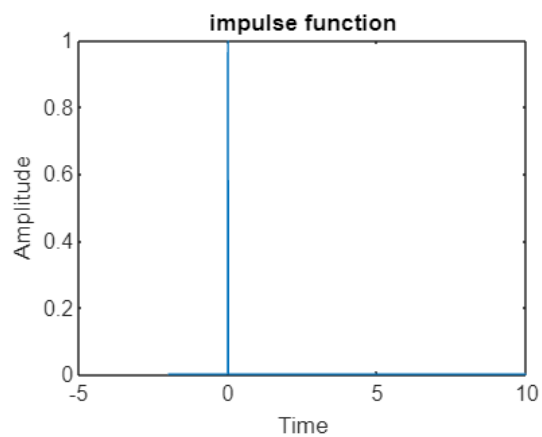
```
title('2D Plot')  
xlabel('Time')  
ylabel('Amplitude')
```

3-D plot

```
clc  
clear all  
t=(-4:0.01:4);  
x=t.^2 ;  
y=4*t ;  
plot3(x,y,t)  
grid on  
xlabel('x-axis')  
ylabel('y-axis')  
zlabel('z-axis')  
title('3D Plot')
```

OUTPUTS:

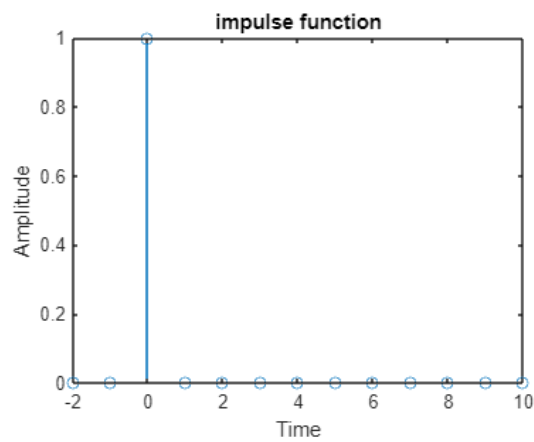
CONTINUOUS TIME SIGNAL PLOTS



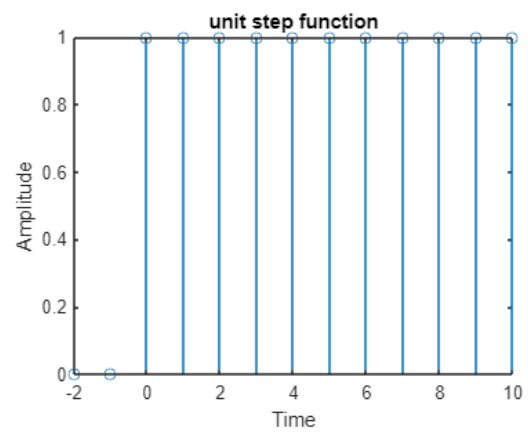
(a) (b) (c) (d) (e)

Figure: Continuous time plots of (a) Impulse Signal, (b) Step signal, (c) Growing and Decaying Exponential signal, (d) Ramp signal, (e) Sinusoidal signal.

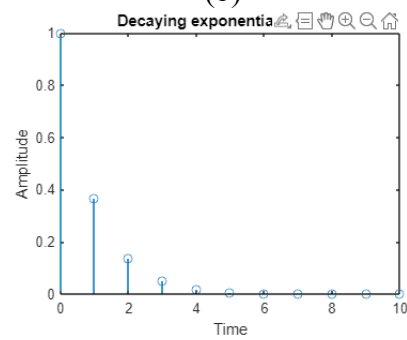
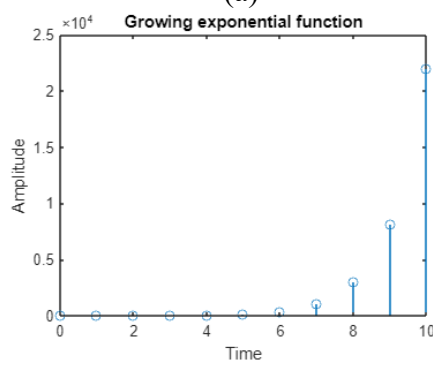
DISCRETE TIME SIGNAL PLOTS



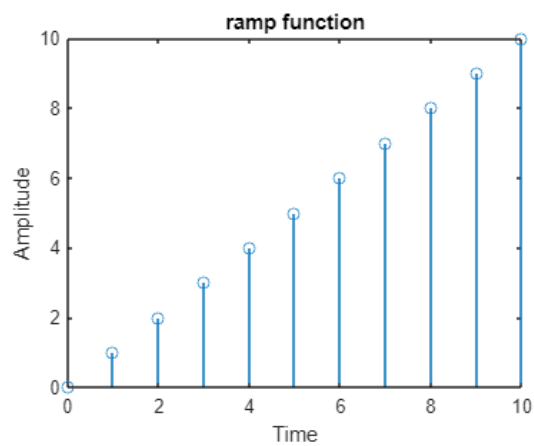
(a)



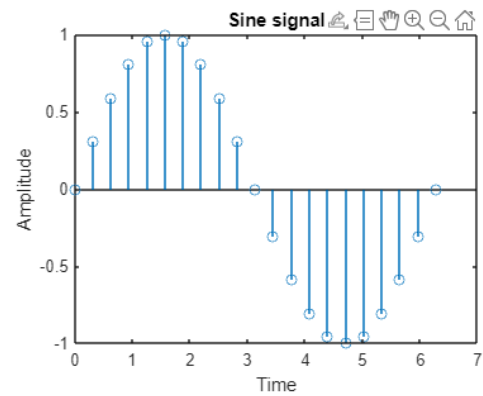
(b)



(c)



(d)



(e)

Figure: Discrete time plots of (a) Impulse Signal, (b) Step signal, (c) Growing and Decaying Exponential signal, (d) Ramp signal, (e) Sinusoidal signal.

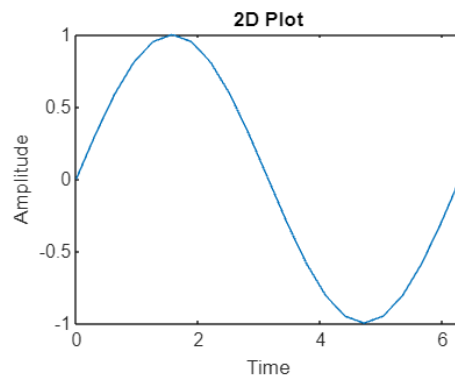


Figure-2 2D Plot of sine signal

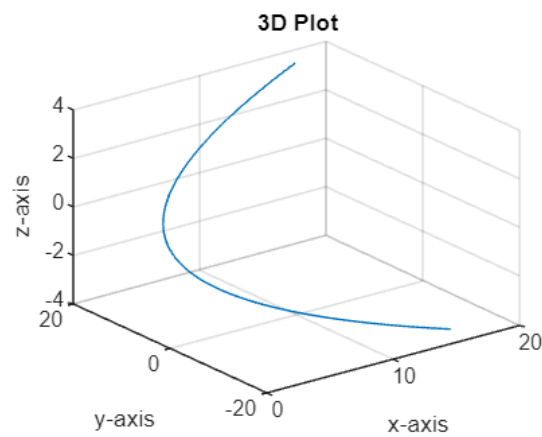


Figure-3 3D plot

Result:

Thus, various MATLAB commands for creating two- and three-dimensional plots are studied, and the continuous and discrete time signals are plotted.

LAB EXPERIMENT 2

AIM: -

To write a MATLAB program to perform amplitude-scaling, time-scaling and time- shifting on a given signal.

SOFTWARE USED: -MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

PROGRAM:

AMPLITUDE SCALING

```
clc
clear all
t=(-3:0.01:7);
unitstep=t>=0 ;
u1=3*unitstep;
ramp = t.*unitstep ;
r1=2*ramp;
plot(t,unitstep)
hold on
plot(t,u1)
hold off
title('Amplitude-Scaling-Step')
xlabel('Time')
ylabel('Amplitude')

figure;
plot(t,ramp)
hold on
plot(t,r1)
hold off
title('Amplitude-Scaling-Ramp')
xlabel('Time')
ylabel('Amplitude')
```

TIME SHIFTING

```
clc
clear all
t=(-1:0.01:10);
impulse=t==0;
unitstep=t>=0;
u1=t>=3;
u2=t>=5;
plot(t,unitstep,'r')
hold on
plot(t,u1,'g')
plot(t,u2,'m')
legend('UnitStep','u1','u2')
title('Time-Shifting')
xlabel('Time')
ylabel('Amplitude')
```

TIME SCALING

```
close all;
clc;
t=-2:0.5:2;
x=zeros(1, length(t));

% unit triangular signal
for i=1:length(t)
    t_val=t(i);
    if t_val>=0 && t_val<=1
        x(i)=1-t_val;
    elseif t_val>=-1 && t_val<=0
        x(i)=1+t_val;
    else
        x(i)=0;
    end
end

plot(t,x);
xlabel('Time');
ylabel('Amplitude');
title('Input Signal');

plot(t.*2,x);
xlabel('Time ');
ylabel('Amplitude');
title('Expanded Signal');

plot(t./2,x);
xlabel('Time ');
ylabel('Amplitude');
title('Compressed Signal');
```

OUTPUT

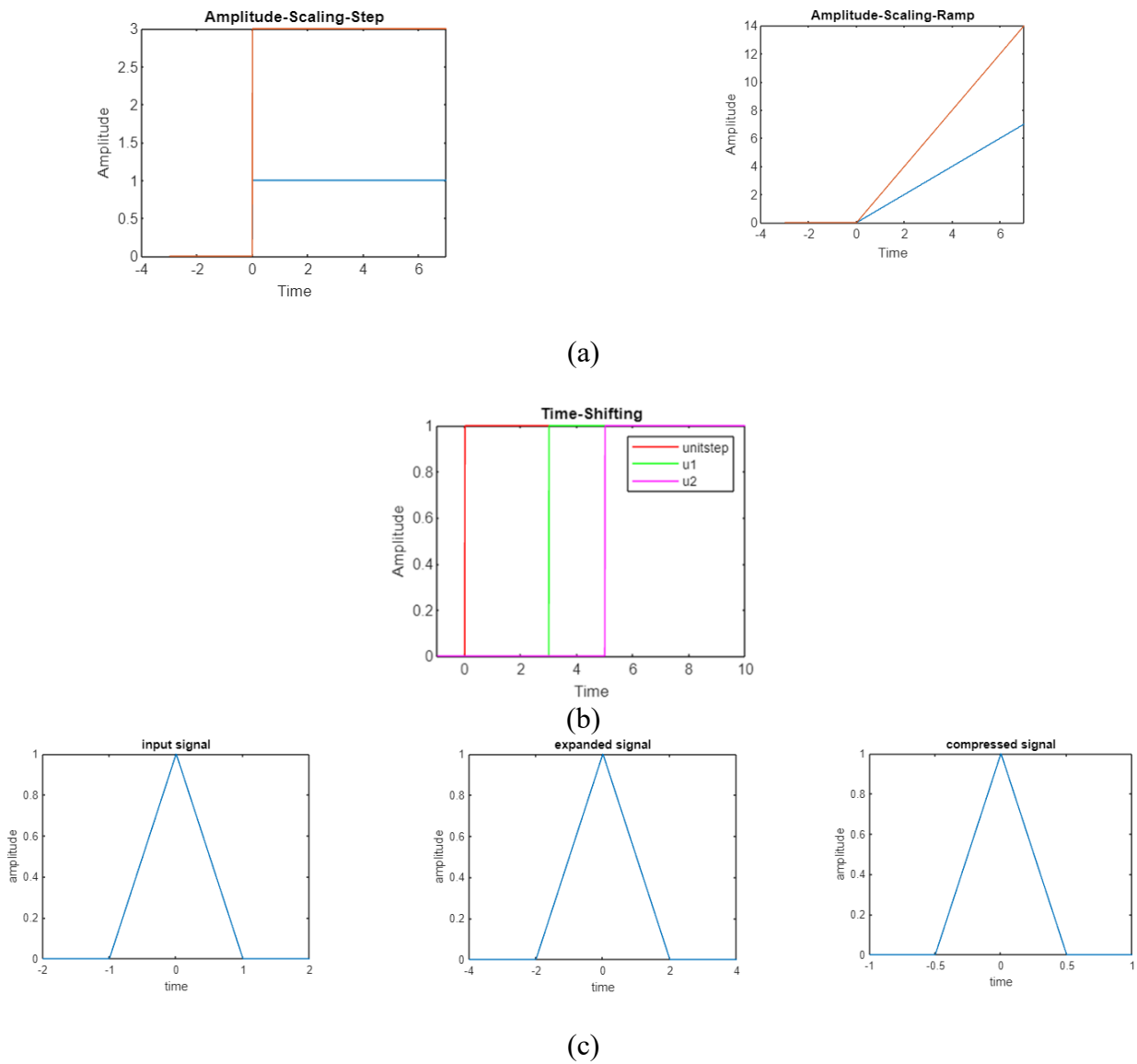


Figure: (a) Amplitude Scaling of sine and ramp signal, (b) Time Shifting, (c) Time Scaling

RESULT: -

A MATLAB program to perform amplitude-scaling, time-scaling and time- shifting on a given signal was successfully implemented and result was obtained.

LAB EXPERIMENT 3

AIM:

To write a MATLAB program to obtain linear convolution of the given sequences.

SOFTWARE USED: MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

PROGRAM:

```
clc;
clear all;
close;
disp('Enter the Length of the First Sequence m = ');
m=input('');

disp('Enter the First Sequence x[m]=');
for i=1:m
    x(i)=input('');
end

disp('Enter the Length of the Second Sequence n = ');
n=input('');

disp('Enter the Second Sequence h[n]=');
for j=1:n
    h(j)=input('');
end
y=conv(x,h);

figure;
subplot(3,1,1);
stem(x);
ylabel('Amplitude');
xlabel('n');
title('x(n) Vs n');

subplot(3,1,2);
stem(h);
ylabel('Amplitude');
xlabel('n');
```

```
title('h(n) Vs n');  
  
subplot(3,1,3);  
stem(y);  
ylabel('Amplitude');  
xlabel('n');  
title('y(n) Vs n');  
disp('Linear Convolution of x[m] and h[n] is y');  
disp(y)
```

INPUT: --

Enter the Length of the First Sequence m =

6

Enter the Length of First Sequence x[m] =

1

2

3

4

5

6

Enter the Length of the Second Sequence n =

6

Enter the Length of Second Sequence h[n] =

1

2

3

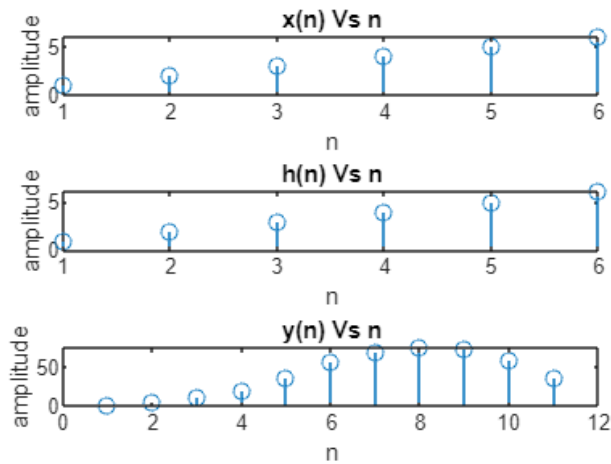
4

5

6

OUTPUT: -

Linear convolution of $x[n]$ and $h[n]$ is $y = 1 \ 4 \ 10 \ 20 \ 35 \ 56 \ 70 \ 76 \ 73 \ 60 \ 36$



RESULT: - Thus, the program for linear convolution is written using MATLAB was successfully implemented and result was obtained.

LAB EXPERIMENT 4

AIM: -

To write a MATLAB program to find the impulse response and step response of a system from its difference equation.

SOFTWARE USED: MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

PROGRAM:

Step Response: Determine the first 50 values of the step response of a system with a known impulse response $h[n]=0.5^n * u[n]$.

```
clc
clear all
N=50;
h=0.5.^[0:49];
x=ones(1,50);
y=conv(x,h);
stem([0:49],y(1:50));
```

Impulse Response: Determine the first 50 values of the step response of a system with a known impulse response $h[n] = 0.5^n$

$*u[n]$. Here define delta as a vector containing 21 elements. Element number 11 is one the rest are zeros.)

```
clc
clear all
delta=[ zeros(1,10), 1, zeros(1,10)];
h=0.5.^[0:49];
y=conv(delta,h);
stem([0:49], y(1:50))
```

OUTPUT:

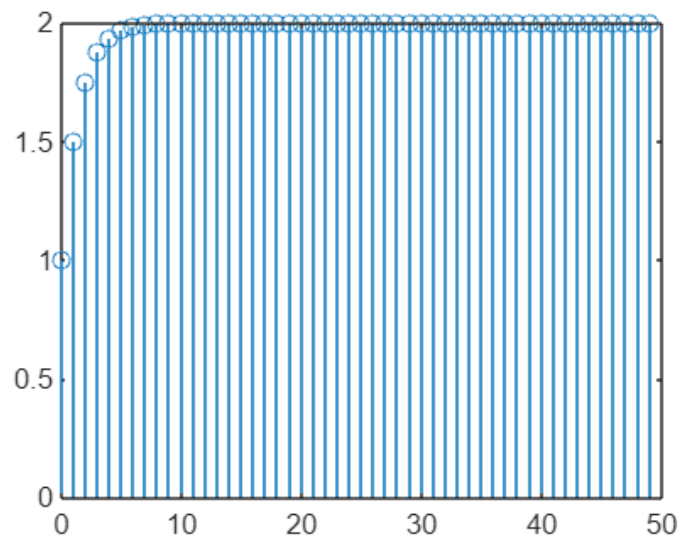


Figure-1: Step Response

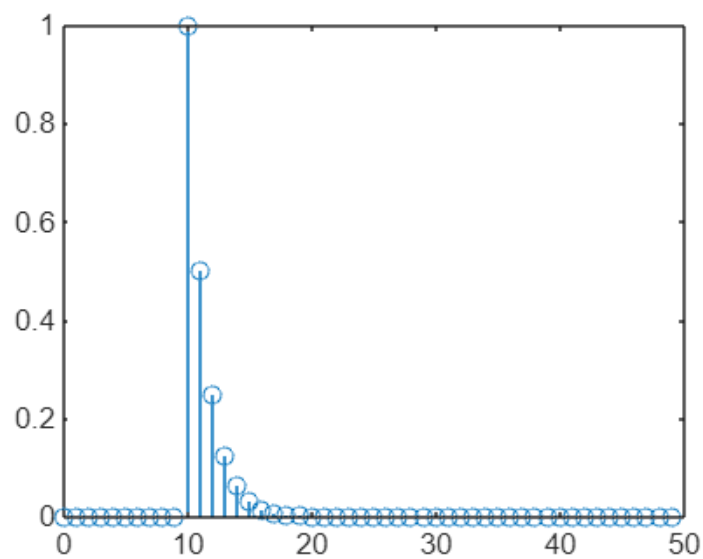


Figure-2: Impulse Response

RESULT:

A MATLAB program to obtain linear convolution of the given sequences was successfully implemented and result was obtained.

LAB EXPERIMENT 5

AIM:

To write a MATLAB program to generate Fourier series of a Square Wave and to calculate the sum of first 10 and 50 terms

SOFTWARE USED: MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

PROGRAM:

```
% Parameters
T = 2*pi;    % Period of the square wave
L = T/2;    % Half period
t = linspace(-T, T, 1000); % Time vector

% Fourier Series coefficients
a0 = 0; % For a square wave, the DC component is zero
N_terms = 10; % Number of terms in the series for the first plot

% Initialize Fourier Series approximation
x_approx_10 = a0 * ones(size(t)); % Initialize with the DC component

% Summing the first 10 terms
for n = 1:2:(2*N_terms-1)
    x_approx_10 = x_approx_10 + (4/(n*pi)) * sin(n * t);
end

% Plot the Fourier series approximation (first 10 terms)
figure;
plot(t, x_approx_10, 'b', 'LineWidth', 1.5);
hold on;
title('Fourier Series Approximation (10 Terms)');
xlabel('Time');
ylabel('Amplitude');
grid on;
hold off;
```

```

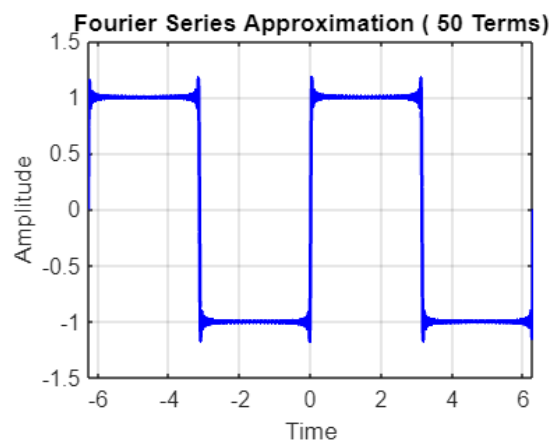
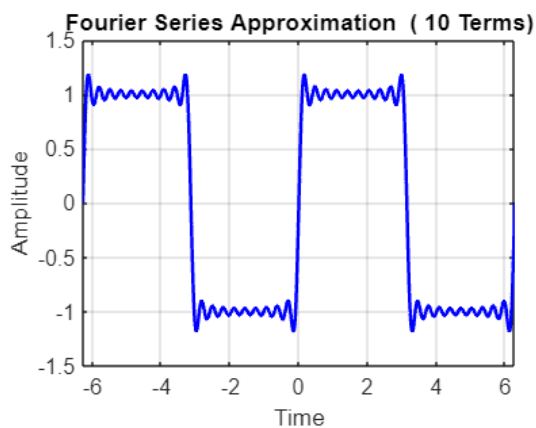
% Sum the first 50 terms and plot the Fourier series as a function of time
N_terms = 50; % Number of terms in the series for the second plot
x_approx_50 = a0 * ones(size(t)); % Initialize with the DC component

for n = 1:2:(2*N_terms-1)
    x_approx_50 = x_approx_50 + (4/(n*pi)) * sin(n * t);
end

% Plot the Fourier series approximation (first 50 terms)
figure;
plot(t, x_approx_50, 'b', 'LineWidth', 1.5);
hold on;
title('Fourier Series Approximation ( 50 Terms)');
xlabel('Time');
ylabel('Amplitude');
grid on;
hold off;

```

OUTPUT:



RESULT:

A MATLAB program to calculate Fourier series and plotting of sum of the first 10 and 50 terms was successfully implemented and result was obtained.

LAB EXPERIMENT 6

AIM:

To write a MATLAB program to calculate and plot using MATLAB Fourier Transform of a given signal.

SOFTWARE USED: MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

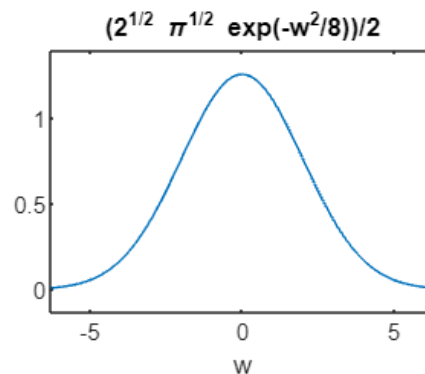
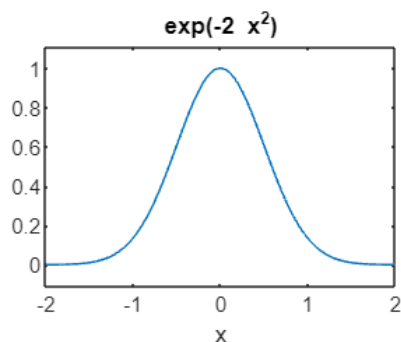
PROGRAM:

Fourier transform of $\exp(-2x^2)$

```
clc
clear all
syms x
f=exp(-2*x^2)           % our function
ezplot(f,[-2,2])        % plot of our function
FT=fourier(f)           % Fourier transform
ezplot(FT)
```

OUTPUT:

Fourier transform of e^{-2x^2} is $\frac{\sqrt{2}}{2} \sqrt{\pi} e^{-\frac{w^2}{8}}$



RESULT:

A MATLAB program to calculate and plot Fourier Transform was successfully implemented and result was obtained.

LAB EXPERIMENT 7

AIM:

To write a MATLAB program to calculate and plot Z-Transform of a given signal.

SOFTWARE USED: MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

PROGRAM:

z transform of $\frac{1}{4^n}$

```
% z transform
clc
clear all
syms z n
a=ztrans(1/4^n)
ezplot(a)
```

Inverse z transform of $\frac{2z}{2z-1}$

```
% Inverse z transform
clc
clear all
syms z n
aa=iztrans(2*z/(2*z-1))
ezplot(aa)
```

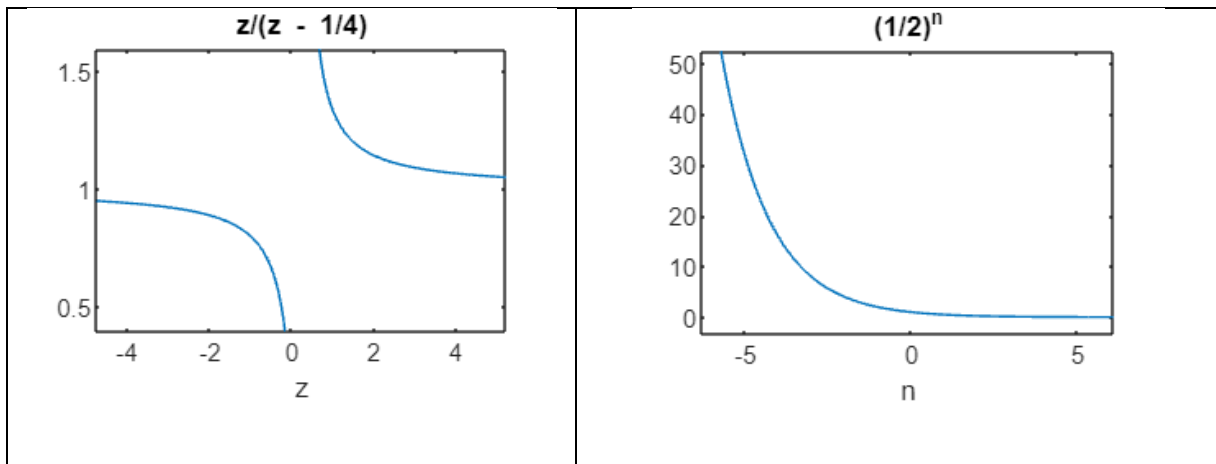
OUTPUT:

Z- Transform of $x(n) = \frac{1}{4^n}u(n)$

$$x(z) = \frac{z}{z - \frac{1}{4}}$$

Inverse Z- Transforms of $x(z) = \frac{2z}{2z-1}$

$$x(n) = \left(\frac{1}{2}\right)^n$$



RESULT:

A MATLAB program to calculate and plot Z-transform was successfully implemented and result was obtained.

LAB EXPERIMENT 8

AIM:

To write a MATLAB program to calculate and plot the pole -zero diagram and ROC of the given system function.

SOFTWARE USED: MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

PROGRAM

For example, if your transfer function is:

$$H(s) = \frac{s^3 + 3s^2 + 3s + 1}{s^2 + 2s + 1}$$

The coefficients in the script would be:

- Numerator: [1 3 3 1]
- Denominator: [1 2 1]

```
% Define the numerator and denominator of the transfer function H(z)
numerator = [1 3 3 1];
denominator = [1 2 1];

% Create a transfer function
H = tf(numerator, denominator, -1); % '-1' specifies that the transfer function is
in z-domain

% Plot the pole-zero diagram
figure;
pzmap(H); % Plot poles and zeros
title('Pole-Zero Diagram');
grid on;

% Calculate poles and zeros
poles = roots(denominator);
zeros = roots(numerator);

% ROC depends on the nature of the system, for illustration, consider:
```

```

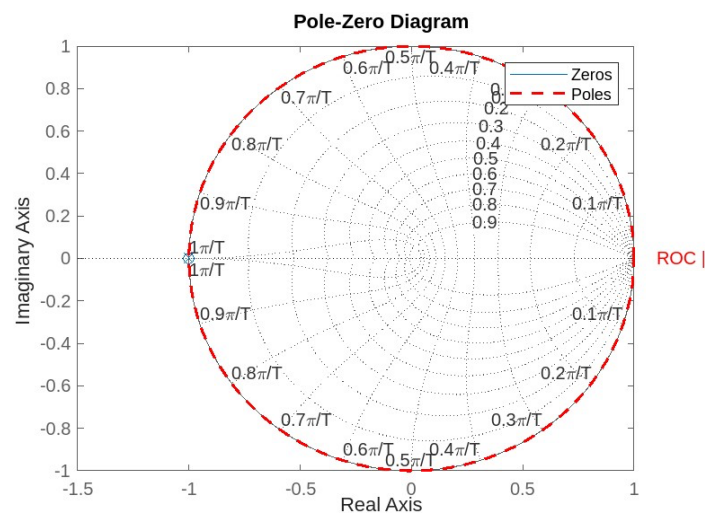
% ROC for a causal system ( $|z| > \max(|\text{poles}|)$ )
roc_radius = max(abs(poles));
theta = linspace(0, 2*pi, 100);
roc_circle = roc_radius * exp(1j * theta);

% Plot the ROC
hold on;
plot(real(roc_circle), imag(roc_circle), '--r', 'LineWidth', 1.5); % ROC circle
legend('Zeros', 'Poles', 'ROC');
hold off;

% Optionally, indicate the ROC on the diagram
text(roc_radius * 1.1, 0, 'ROC  $|z| > \max(|\text{poles}|)$ ', 'Color', 'red');

```

OUTPUT:



```

Poles:
    -1
    -1
Zeros:
    -1.0000 + 0.0000i
    -1.0000 - 0.0000i
    -1.0000 + 0.0000i

```

RESULT:

A MATLAB program to calculate and plot the pole -zero diagram and ROC of the given system function was successfully implemented, and result was obtained.

LAB EXPERIMENT 9

AIM:

To write a MATLAB program to plot magnitude and phase response of a given system.

SOFTWARE USED: MATLAB R2018a

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

PROGRAM:

```
clc
clear all
close all
t = 0:1/100:10-1/100;
x = sin(2*pi*15*t) + sin(2*pi*40*t);
y = fft(x);
m = abs(y);
p = unwrap(angle(y));
f = (0:length(y)-1)'*100/length(y);

subplot(2,1,1)
plot(f,m)
ylabel('Abs. Magnitude')
grid on

subplot(2,1,2)
plot(f,p*180/pi)
ylabel('Phase [Degrees]')
grid on
xlabel('Frequency [Hertz]')
```

OUTPUT:

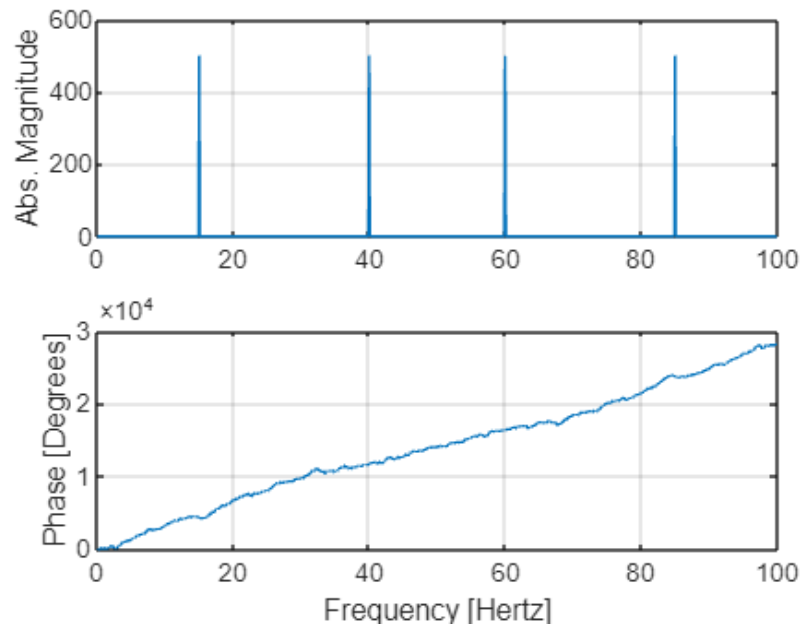


Figure-1

RESULT:

A MATLAB program to plot magnitude and phase response of a given system was successfully implemented and result was obtained.

LAB EXPERIMENT 10

AIM:

To design a model to amplify a sine wave by a factor of two and a model to check Linearity/Non-Linearity of a system using SIMULINK.

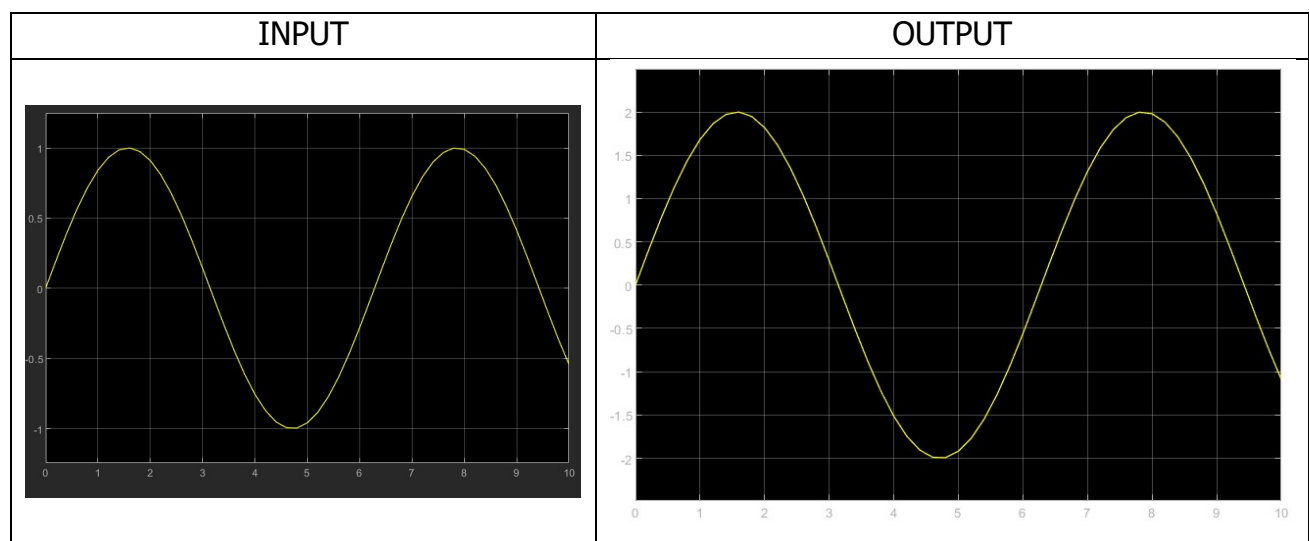
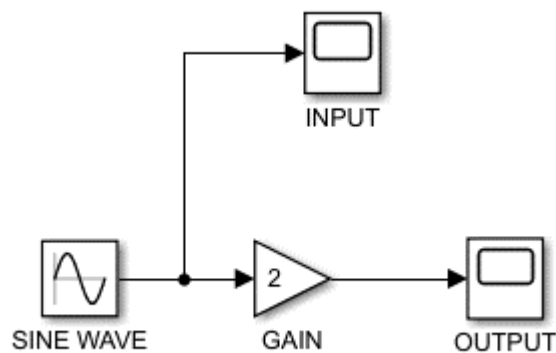
SOFTWARE USED: MATLAB R2018a (SIMULINK)

PROCEDURE: -

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

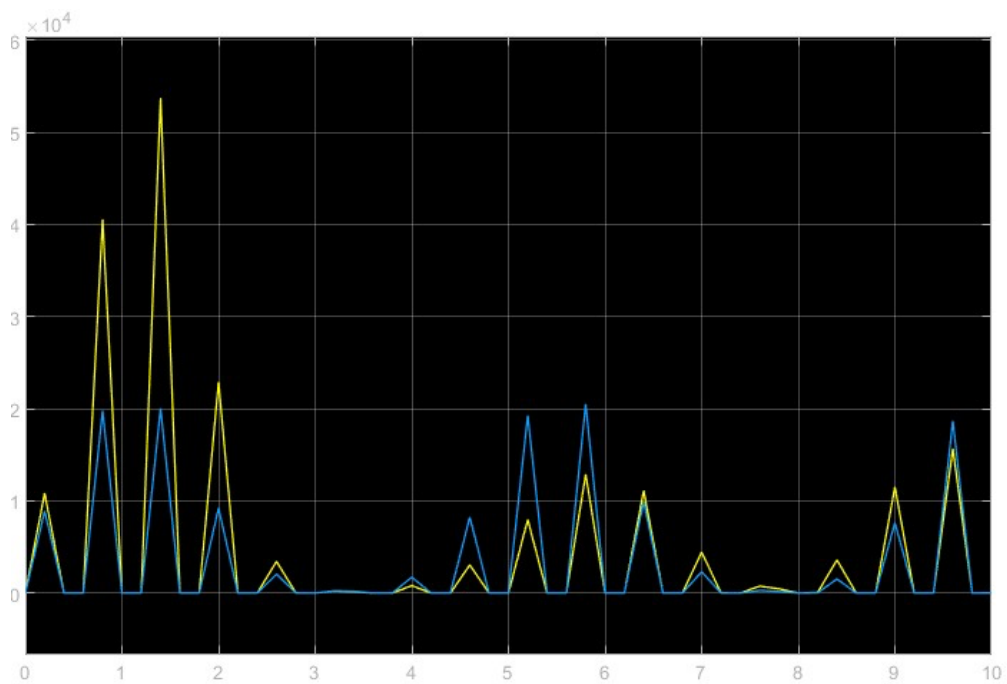
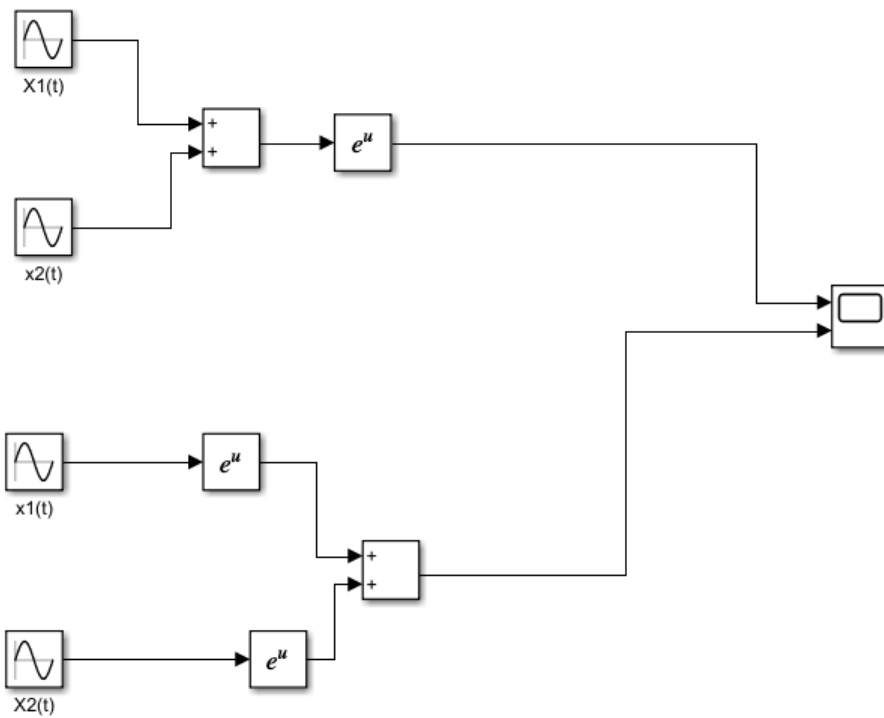
MODEL

AMPLIFICATION



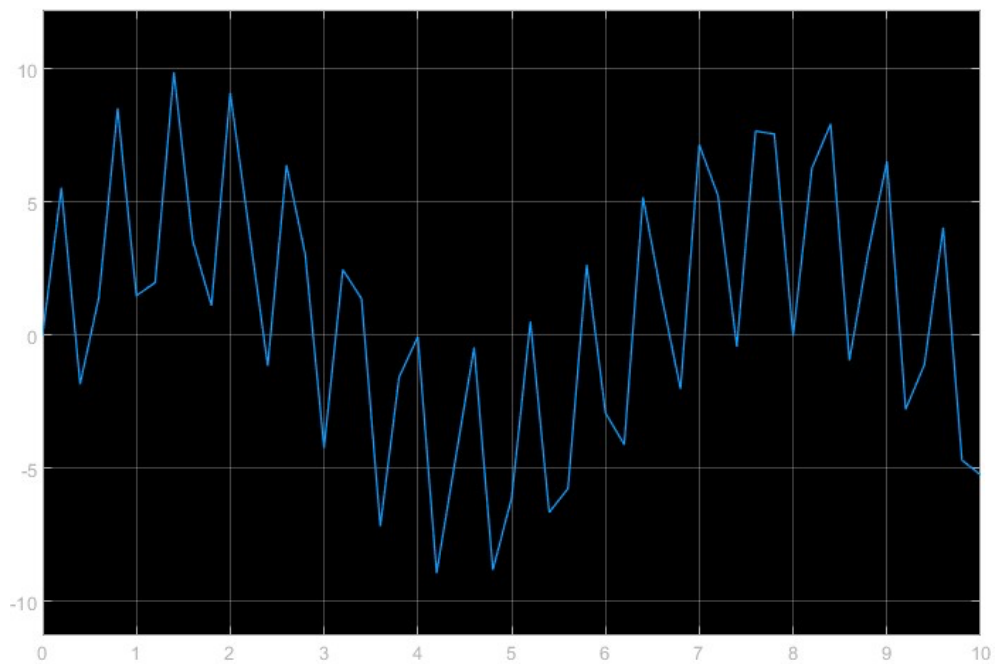
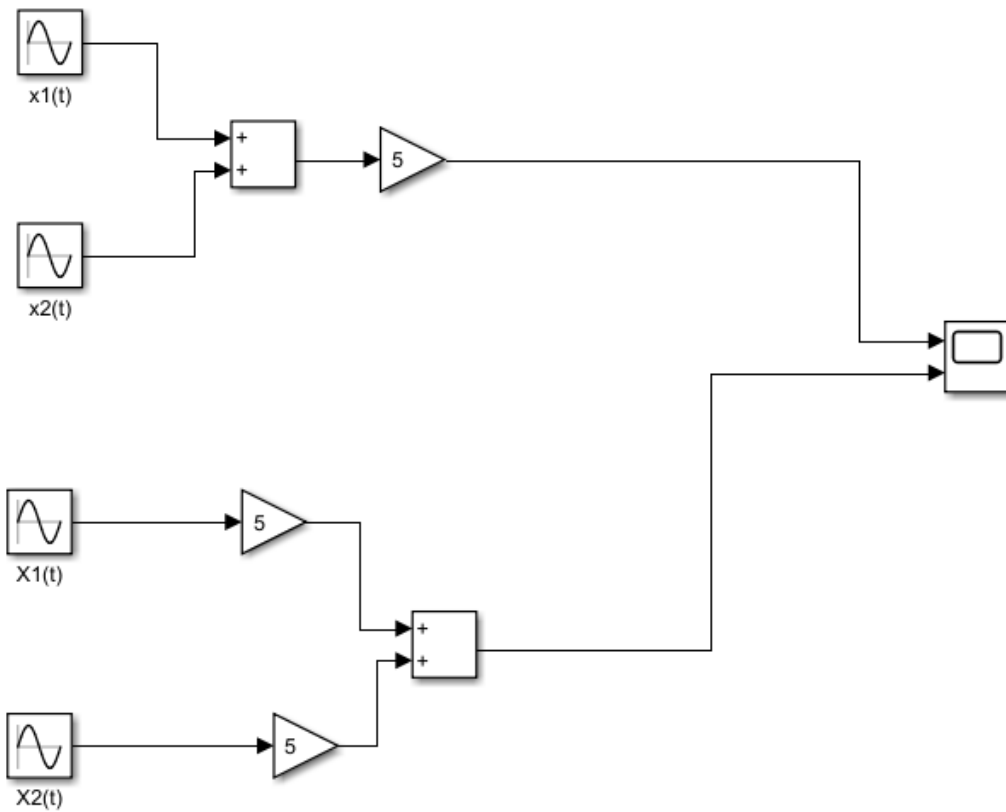
Non-Linear system

$$e^{(\sin 10t + \sin t)} \neq e^{\sin 10t} + e^{\sin t}$$



Linear system

$$5\sin t + 5\sin 10t = 5(\sin t + \sin 10t)$$



RESULT:

A SIMULINK model to amplify sine signal and a model to check Linearity/Non-Linearity of a system was successfully implemented and result was obtained.